

Complete Python Mastery Roadmap

This roadmap is designed to take a learner from Beginner to Advanced level in Python programming.

1. Python Basics

• Syntax
• Variables
• Data Types
• Input / Output
• Operators
• Type Conversion
• Comments

2. Control Flow

• if / else / elif
• Nested Conditions
• for loop
• while loop
• break / continue / pass

3. Functions

• Function Declaration
• Parameters
• Return Values
• Default Arguments
• *args
• **kwargs
• Lambda Functions

- Recursion

4. Data Structures

- List
- Tuple
- Set
- Dictionary
- List Comprehension

5. Strings

- String Methods
- Formatting
- f-strings
- Slicing
- replace()
- split()
- join()

6. File Handling

- open()
- read()
- write()
- append()
- File Modes

7. Exception Handling

- try
- except

- finally
- Custom Exceptions

8. Modules & Packages

- import
- Built-in Modules
- Custom Modules
- pip

9. Object Oriented Programming (OOP)

- Class
- Object
- Constructor
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction

10. Advanced Python

- Decorators
- Generators
- Iterators
- Context Managers
- Closures
- map()
- filter()
- reduce()

11. Python Libraries

- NumPy
- Pandas
- Matplotlib
- Scikit-learn
- TensorFlow
- PyTorch

12. Database

- SQLite
- MySQL
- PostgreSQL
- ORM Basics

13. APIs

- REST API
- requests library
- JSON Handling

14. Automation & Scripting

- Web Scraping
- Automation Scripts
- Bots

15. Testing

- unittest
- pytest

16. Concurrency

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| • Threading |
| • Multiprocessing |
| • Asyncio |

17. DSA in Python

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| • Arrays |
| • Linked Lists |
| • Stacks |
| • Queues |
| • Trees |
| • Graphs |
| • Sorting |
| • Searching |

18. Projects

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|-------------------|
| • Calculator App |
| • To-do App |
| • Chatbot |
| • AI Project |
| • Automation Tool |
| • Web App |

19. Professional Skills

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| • Git & GitHub |
| • Linux Basics |
| • Debugging |
| • Clean Code |

- Problem Solving

Goal: After completing this roadmap, you will be capable of building professional Python applications, AI systems, automation tools, APIs, and advanced software projects.